

Requirements Gathering

SOC Automation Lab for Threat Detection and Incident Response

Stakeholder Analysis

Stakeholder	Role	Needs
SOC Analyst	Monitor alerts and investigate incidents	Access logs and alerts for investigation
Security Administrator	Manage security infrastructure	Configure monitoring systems and rules
System Administrator	Maintain servers and systems	Ensure system stability and performance
Organization	Ensure security protection	Reliable threat detection and response

User Stories & Use Cases

- As a SOC analyst, I want to monitor alerts so that I can detect suspicious activity quickly.
- As a security administrator, I want to configure security tools so that the system can detect threats effectively.
- As a SOC analyst, I want to investigate incidents so that I can determine whether an attack occurred.
- As a system administrator, I want to monitor system performance so that the SOC infrastructure operates reliably.

Functional Requirements

Requirement	Description
Log Collection	Collect logs from endpoints and servers
Event Monitoring	Continuously monitor system events
Threat Detection	Detect suspicious behavior using rules
Alert Generation	Generate alerts when threats are detected
Incident Management	Create and track security incidents
Threat Intelligence Integration	Enrich alerts using threat intelligence
Attack Simulation	Simulate attacks to test detection capabilities

Non-Functional Requirements

Category	Requirement
Performance	System should process logs in near real-time
Security	Access restricted to authorized users

Reliability	System should operate continuously
Scalability	Support additional monitored endpoints
Usability	Interface should be easy for SOC analysts
Availability	System should maintain high uptime